

## A need for standards

“It becomes very interesting for us to witness this seamlessly connected virtual world as part of the Metaverse. It opens the door to a wide number of possibilities and we have yet to see how far we can truly go with it. It is important to adopt at scale; otherwise, it all falls flat. We have seen some of the standards that ensure interoperability, such as Razer’s open source virtual reality (OSVR). We have seen that it encourages experimentation with the development of headsets as well. But if you look at the flip side, on the device layer you will see the OpenXR standard being badly adopted by manufacturers,” says Vinay Rajagopal, ISV technology lead architect for partner ecosystem at Red Hat.

“Encouraging the development of a decentralised, distributed and interoperable virtual world built on the foundation of open standards also creates multiple opportunities. We have already started seeing a proliferation of 3D environments that have been created by the users themselves. This is where the blockchain has become imperative for the details surrounding a transaction and the digital assets or the avatars. It has become important for us to put these assets on multiple ledgers, while making sure that you understand who has built, sold or purchased a particular digital asset. This is important from a legal aspect too,” he adds.

“Open source is the ground zero for the development of technologies like the Metaverse. All the biggest projects in the last five to ten years that have been able to garner the most attraction are the ones that are open sourced — artificial intelligence, blockchain, and containerisation, to name a few. Open source comes with a great number of benefits. It allows you to collaborate with the smartest minds from every corner of the world; your work gets constantly reviewed and used, and there is a constant feedback loop,” says Gandhali Samant, director, cloud



**Dr Lokesh Boregowda**, director, Samsung R&D Institute India



**Vinay Rajagopal**, ISV technology lead architect for partner ecosystem at Red Hat.



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**Dr Sumit Kalra**, assistant professor in the Department of Computer Science and Engineering, Indian Institute of Technology, Jodhpur

solution architecture, digital and app innovation, Microsoft, India. “A good aspect of working with open source is that there will always be components ready to be used, which saves a lot of time as you don’t need to start from scratch. As the Metaverse continues to gain traction, more tools and platforms will appear to assist the developers.”

“Even though open source comes with a lot of benefits such as quick development and large communities, it is important to note these communities own the software itself. In 2014, a bug was detected in Unison. Its source code had been open for so long that it made it hard to estimate the time hackers had been exploiting it,” says Dr Sumit Kalra, assistant professor in the Department

of Computer Science and Engineering, Indian Institute of Technology, Jodhpur. “Today whatever we do with the Metaverse is going to have a tremendous impact on the future of its development. If we are going to continue Metaverse development with open source then it is crucial for us to adopt open standards.”

## The right hardware

Syed Mohammed Khasim, senior manager, software development - Sitara ARM MPU, Texas Instruments, notes: “In the last couple of decades I have seen the evolution of open source, starting from Linux. It all essentially started for shared software development. The cost of developing software turns out to be a big challenge